

Testing Document - PAMS

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Part 1 - Automated Testing

I used Python unittest to write 52 automated unit tests. The test file is tests/test_pams.py. To run the tests:

```
python -m pytest tests/test_pams.py -v
```

All 52 tests passed. The table below shows what each test class covers.

Test Class	No. of Tests	What It Tests
TestTenantModel	5	Checks tenant validation works - NI number format, name, email, phone
TestLeaseModel	4	Checks lease logic - days remaining, 5% early exit penalty, active status
TestInvoiceModel	5	Checks invoice logic - overdue detection, mark as paid, receipt generation
TestApartmentModel	2	Checks the occupancy status method returns correct value
TestMaintenanceModel	2	Checks the is_resolved method works correctly
TestAuthService	5	Checks login accepts correct credentials and rejects wrong ones
TestTenantService	6	Checks create, update, delete, search, and duplicate NI rejection
TestApartmentService	4	Checks create apartment, assign tenant, vacate, filter available
TestFinanceService	4	Checks invoice creation, recording payment, marking overdue, summary
TestMaintenanceService	3	Checks create request, resolve request, cost summary
TestReportService	3	Checks occupancy report, maintenance cost report, complaint log
TestNotificationService	2	Checks sending a notification and marking it as read
TestSecurity	3	Checks password is stored as hash not plain text
TestUserService	4	Checks create user, deactivate, filter by location
Total	52	All 52 tests passed

Part 2 - Manual Testing

I ran the application and tested each feature manually. Below are the results.

Login

N o.	Test	Input	Expected	Actual	Pass/Fail
1	Login with correct admin details	admin1@pams.com / admin123	Dashboard opens with all pages	Worked, all 7 pages showed in sidebar	Pass
2	Login as front desk staff	frontdesk1@pams.com / desk123	Only Tenants, Complaints, Maintenance show	Correct, Finance and Reports not visible	Pass
3	Wrong password	admin1@pams.com / wrongpass	Error message, no login	Said incorrect email or password	Pass
4	Both fields left empty	Email and password blank	Tells user to fill in fields	Message appeared asking to enter details	Pass
5	Email that does not exist	fake@test.com / anything	Login fails	Error message showed	Pass
6	Check password is not plain text	Looked at database directly	Password should be a hash	64 character SHA-256 hash confirmed	Pass

Tenant Management

N o.	Test	Input	Expected	Actual	Pass/Fail
7	Add a new tenant	NI: AB999999Z, Name: Test User Email: test@test.com	Tenant added to the list	Tenant appeared after saving	Pass
8	Add tenant with NI already used	NI: AB111111A (already in system)	Error - NI already registered	Error showed, tenant not added	Pass
9	Email with no @ symbol	Email: notanemail	Form shows error	Said valid email is required	Pass
10	Leave name field empty	Name field blank	Form shows error	Said name is required	Pass
11	NI number too short	NI: AB1	Form shows error	Said minimum 9 characters	Pass
12	Search tenant by name	Typed Ahmed in search bar	Only Ahmed Hassan shows	Table filtered to one result	Pass

N o.	Test	Input	Expected	Actual	Pass/Fail
13	Edit tenant phone number	Changed phone for existing tenant	New number saved	Updated and visible in table	Pass
14	Early exit - active lease (£1200)	Clicked Early Exit on Imene Benali	Penalty £60 shown, lease terminated	Penalty correct, lease ended, apartment freed	Pass
15	Early exit - no active lease	Clicked Early Exit on tenant with no lease	Warning message	Warning showed correctly	Pass

Apartments

N o.	Test	Input	Expected	Actual	Pass/Fail
16	Add a new apartment	Unit Z999, Manchester, studio, £750	Shows in list as available	Appeared with correct details	Pass
17	Assign tenant to apartment	Linked tenant to free apartment	Status changes to Occupied	Updated correctly in table	Pass
18	Filter to available only	Used available filter	Only free apartments show	Filter worked	Pass
19	Filter by city - London	Selected London from dropdown	Only B101 and B201 show	Correct apartments showed	Pass

Finance

N o.	Test	Input	Expected	Actual	Pass/Fail
20	Create an invoice	Tenant: Ahmed, Amount: £950	Invoice shows as unpaid	Shown in All Invoices tab	Pass
21	Record a payment	Clicked Record Payment on invoice	Invoice changes to paid	Status changed to PAID	Pass
22	Check overdue invoices	Invoice with past due date	Shows in Overdue tab	Tab populated correctly	Pass
23	Send late payment alerts	Clicked Send Late Alerts button	Notifications created	Confirmation message showed	Pass
24	Check financial summary	Mix of paid and unpaid invoices in DB	Cards show correct totals	Figures matched invoice data	Pass

Maintenance

N o.	Test	Input	Expected	Actual	Pass/Fail
25	Log a maintenance request	Tenant Ahmed, Issue: Leaking tap	Request appears as open	Shown with correct details	Pass
26	Submit with no description	Left description blank	Error shown	Said description is required	Pass
27	Resolve a request	Notes: Fixed tap, 1.5 hrs, £45	Status changes to resolved	Updated, notification sent	Pass
28	Filter by open status	Set filter dropdown to open	Only open requests show	Filter worked correctly	Pass

N o.	Test	Input	Expected	Actual	Pass/Fail
29	Maintenance cannot see Finance	Logged in as maintenance1@pams.com	Finance not in the menu	Finance page was not there	Pass

Complaints

N o.	Test	Input	Expected	Actual	Pass/Fail
30	Log a complaint	Jack Thompson - Noisy neighbours	Complaint shows as open	Showed in the table	Pass
31	Submit with no description	Left description blank	Error shown	Error appeared	Pass
32	Resolve a complaint	Entered resolution notes	Status changes to resolved	Updated, notes visible in table	Pass
33	Filter by open	Set filter to open	Only open complaints show	Filtered correctly	Pass
34	Stats cards update	Resolved one open complaint	Open count drops, resolved goes up	Cards updated after resolving	Pass

Reports

N o.	Test	Input	Expected	Actual	Pass/Fail
35	Occupancy report - all cities	No location filter applied	All 4 cities listed	Bristol, Cardiff, London, Manchester showed	Pass
36	Occupancy report - Bristol	Selected Bristol from filter	Only Bristol data shows	Correct	Pass
37	Financial summary report	Opened financial report	Shows collected vs pending rent	Matched the invoice data	Pass
38	Maintenance cost report	Opened maintenance cost report	Shows resolved requests with costs	Boiler repair £180, 3.5hrs showed	Pass

User Management and Security

N o.	Test	Input	Expected	Actual	Pass/Fail
39	Admin creates new staff	Name: New Staff, Role: front desk	Account created, staff can log in	User in table, login worked	Pass
40	Deactivate a user account	Clicked Deactivate on a user	User cannot log in anymore	Login failed after deactivation	Pass
41	Admin sees only own city	Logged in as Bristol admin	Only Bristol users show	London users not shown	Pass
42	Manager adds new city	City: Edinburgh, Address: 10 Royal Mile	City appears in list	Edinburgh showed in city table	Pass

N o.	Test	Input	Expected	Actual	Pass/Fail
43	Duplicate city rejected	Tried to add Bristol again	Error - city already exists	Error showed, not added again	Pass
44	SQL injection in login	Email: admin' OR '1'='1	Login should fail	Rejected, no access granted	Pass
45	SQL injection in search	'; DROP TABLE tenants; --	Nothing should happen	No results, table still there	Pass
46	Finance cannot see Tenants	Logged in as finance1@pams.com	Tenant page not in menu	Tenant page was not there	Pass

Automated testing: 52 unit tests, all passed. Manual testing: 46 test cases, all passed. Total: 98 tests.